

Probability & Statistics

(MT2005)

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Course Instructor(s)

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MID-2 Exam

Total Time (Hrs): 1

Total Marks: 45

Total Questions: 2

ALL SECTIONS BCS

Roll No	Section	Student Signature
Do not write below this line		

- Attempt all questions.
- Lead Pencil work wouldn't be marked or claimed for rechecking. Use permanent ink pen.
- If you found any ambiguity in the data then do not ask anything to the invigilator, just make an assumption and continue solving your paper.
- Clearly mention the formulas required to answer Question No. 2.

CLO 2: Probability and Distributional Analysis: Apply foundational principals of probability to analyze experiments, including Bayes' theorem, evaluate discrete and continuous distributions, and explore applications in machine learning.

Q1.

[Marks= 25+5]

Keyword-Based Spam Email Filtering: A Case Study of University Email Systems

To safeguard its official email communication channels, a university put in place an anti-spam software system. The project's objective was to automatically identify and filter spam emails so that only valid emails are sent to students, instructors, and administrative personnel. A dataset of emails was gathered during the project and classified as either Ham (legitimate) or spam. The system was implemented on the university's email servers following training and assessment. Consequently, the automated filtering system greatly decreased the quantity of spam emails and enhanced the dependability and effectiveness of communication throughout the university.

For simplicity, suppose the spam detection model uses four important keywords that frequently appear in emails:

free, prize, offer, unsubscribe

From a dataset of 200 emails, it was observed that 160 emails were Ham and 40 emails were Spam. Further analysis showed the following patterns:

- The word "free" appeared in 28 spam emails and 16 ham emails.
- The word "prize" appeared in 24 spam emails and 8 ham emails.
- The word "offer" appeared in 32 spam emails and 20 ham emails.
- The word "unsubscribe" appeared in 36 spam emails and 10 ham emails.

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Now suppose two new emails arrive:

- i. **Email 1:**
"Congratulations! You have received a **free prize offer**. If you do not wish to receive it, unsubscribe here."
- ii. **Email 2:**
"Students can get a **free online course** this week. If you are interested, check the special **offer** available now."
 - a) Compute the posterior probabilities and classify each email (Email 1 and Email 2) separately as Spam or Ham."
 - b) Suppose three different Machine Learning Classifiers independently detect an incoming spam email with probabilities of 0.2, 0.3, and 0.5, respectively. The system received a spam email, what is the probability that at least one classifier detects it as spam? Assume that the classifiers operate independently.

CLO 2: Probability and Distributional Analysis: Apply foundational principals of probability to analyze experiments, including Bayes' theorem, evaluate discrete and continuous distributions, and explore applications in machine learning.

Q2:

[Marks=15]

Sr.	SCENARIOS	RESPONSE
1	A machine learning engineering team is studying the relationship between training time (X) measured in seconds and model performance (Y) measured as accuracy (in proportion) across multiple experiments. The following summary statistics are observed: $\sigma_X = 3.2$, $\sigma_Y = 0.60$, $\sigma_{XY} = 1.10$ Determine the correlation coefficient between X and Y.	
2	A system selects 2 packets without replacement from a total of N packets, where: • a = number of high-priority packets • b = number of error packets • c = number of normal packets Let 'X' and 'Y' represent number of high-priority packets selected and number of error packets selected. Construct a joint Probability Function $f(x, y)$.	
3	Consider a newspaper circulation department where it is known that 84% of the households in a particular neighborhood subscribe to the daily edition of the paper. In addition, it is known that the probability that a household which already holds a daily subscription also subscribes to the Sunday edition is 0.75. What is the probability that a household subscribes to both the daily and Sunday editions of the newspaper?	

4	Two students scheduled interviews for summer employment at an Institute. In each case the interview results in either an offer for a position or no offer. Experimental outcomes are defined in terms of the results of the two interviews. a. List the experimental outcomes. b. Define a random variable that represents the number of offers made. Is the random variable discrete or continuous?	
5	A continuous RV 'X' has PDF $f(x) = e^x$, $0 < x < 1$, else 0. Check if the given function is a valid PDF or not.	
6	A random variable Y has PMF: $P(Y = 1) = 0.2$, $P(Y = 2) = 0.5$, $P(Y = 3) = 0.3$. Find the mean value of $g(x) = 2Y+1$, if $E(Y) = 2.1$.	
7	In a factory, Machine A produces 40% of the total items, of which 3% are defective. Machine B produces the remaining 60% of the items, of which 4% are defective. Are the events 'an item is defective' and 'the item was produced by Machine A' independent? Justify your answer numerically.	
8	A system generates passwords using six different characters: A, B, C, D, E, and F. Each password must use all characters exactly once, with the restriction that A and B must not appear consecutively. How many such valid passwords can be formed?	

Handwritten notes and scribbles, including a circled '2' and some illegible markings.

Handwritten diagram showing a sequence of characters with arrows indicating constraints or transitions between them.